

Multi-Cancer Early Detection Screening Report

Next-Generation Sequencing cfDNA Methylation Profiling (Synthetic Demonstration)

Patient Demographics	Ordering Clinician	Specimen Chain of Custody
Name: Jane Q. Sample DOB: 05/12/1972 (Age 54) Sex: Female Patient ID: P-44556677	Provider: Dr. Avery Sterling, MD Facility: Aurora Health Institute NPI: 1234567890 Location: Silver City, NM	Specimen ID: SYN-992834-X Specimen Type: Blood / Plasma Tube: Streck cfDNA BCT (10.0 mL) Collection Date: Aug 7, 2026 Received Date: Aug 8, 2026 Report Date: Aug 14, 2026

Result: Cancer Signal Detected

Status: Final
Assay: cfDNA Bisulfite NGS

Test Result Summary: A cancer signal was detected in this synthetic blood sample. This result indicates that cell-free DNA (cfDNA) methylation patterns associated with cancer were identified. Further clinical evaluation, such as imaging or other diagnostic tests, is required to confirm the presence of cancer and determine the location.

Laboratory Observations & Biomarkers

Test / Analyte	LOINC	Result	Reference Range	Interpretation
Cancer Signal Status	94076-7	Cancer Signal Detected	Cancer Signal Not Detected	Abnormal (A)
Predicted Cancer Signal Origin 1	94077-5	Lung (85%)	None	Abnormal (A)
Predicted Cancer Signal Origin 2	94077-5	Pancreas (12%)	None	Abnormal (A)
cfDNA Concentration	Local	14.2 ng/mL	5.0 - 25.0 ng/mL	Normal
Sequencing Coverage Depth	Local	32,500x	>= 20,000x	Normal
Sequencing QC Pass Rate	Local	99.7%	>= 95.0%	Normal

Clinical Interpretation & Recommended Next Steps

The 'Cancer Signal Detected' result is not a diagnosis of cancer. It means that genomic signals often associated with malignancy were found. The predicted origins (Lung 85% and Pancreas 12%) should guide subsequent diagnostic workup. Recommended next steps include:

- High-resolution imaging:** Low-dose or contrast-enhanced chest CT and dedicated abdominal MRI / endoscopic ultrasound (EUS).
- Specialist consultation:** Prompt referral to thoracic oncology and gastroenterology / surgical oncology teams.
- Diagnostic tissue biopsy:** If a suspicious lesion is identified on diagnostic imaging.

Methodology & Limitations

Methodology: Plasma cfDNA undergoes targeted bisulfite conversion followed by deep Next-Generation Sequencing (NGS) on Illumina NovaSeq 6000 systems. Epigenetic methylation patterns are classified using proprietary machine-learning algorithms calibrated across 50+ cancer tissue types.

Limitations: This test is an early detection screening tool and is not a definitive histological diagnosis. A negative result does not completely exclude malignancy.

Laboratory Director Authorization: Dr. Eleanor Hayes, MD, PhD, FCAP | Electronically Signed | Date: Aug 14, 2026 | CLIA ID: 00D1234567